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Level measurement and influencing factor analysis of agricultural sustainable development in Shandong Province, China

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Abstract
Agricultural sustainable development is vital due to its direct impact on food security, ecological security, and social stability. This study measured and analyzed the Agricultural Sustainable Development Level (ASDL) across Shandong Province and quantified its key drivers. Results demonstrate a clear spatiotemporal evolution pattern, with ASDL showing overall improvement alongside significant ranking changes and distinct regional variations. The XGBoost-SHAP framework identified key internal drivers: primary industry share, rural residents' disposable income, afforested area, and mechanical power per unit area. Major external factors included population density, GDP per capita, fiscal revenue scale, and ventilation coefficient. Among these factors, primary industry share, rural income, afforested area, and fiscal revenue scale positively influenced ASDL, while mechanical power intensity and GDP per capita negatively influenced it. Population density and ventilation coefficient exhibited significant threshold effects. These findings offer empirical support for formulating targeted policies to enhance agricultural sustainable development in Shandong Province.

Keywords Agricultural economy · Agricultural resources and environment · Agricultural sustainable development · XGBoost model · SHAP method

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1 Introduction

As China's economy enters a new stage of development, promoting green transformation and sustainable development has become an urgent task. The traditional extensive growth model has severely damaged natural resources and ecosystems while hindering sustainable economic and social development. In this context, sustainable development theory advocates for coordinated unity among economic, social, and environmental systems. This concept provides core theoretical support for tackling development challenges. Advancing ecological civilization and achieving green sustainable development are critical for China and the world in the context of the new normal in economic development.

As a fundamental sector of the national economy, agricultural sustainability is directly tied to food security, ecological security, and social stability. However, agricultural modernization has relied heavily on chemicals like pesticides, fertilizers, and agricultural films, coupled with widespread machinery use. While boosting productivity, this has led to severe issues, including soil pollution, biodiversity loss, and land degradation (Moinet et al., 2025). This trend partially aligns with the early stages of the Environmental Kuznets Curve (EKC), where environmental pressure rises with agricultural output (Hsieh & Gramig, 2023). Studies indicate that non-point source agricultural pollution has become a major driver of environmental degradation (Azam et al., 2024). Against this backdrop, reconciling agricultural development with ecological protection, mitigating its negative ecosystem impacts, and achieving sustainable agriculture have become pressing issues in agricultural and environmental economics.

In 2023, the No. 1 Central Document explicitly identified "agricultural green transformation" as a distinct clause, supported by special central government funding. It set a target to increase the coverage rate of green pest control by 10% points by 2025, underscoring its strategic priority at the national level. Shandong Province, a benchmark for modern agricultural development in China, ranks first nationally in key indicators including total agricultural output value, the value of processed agricultural products, and export scale. It also plays a critical role in ensuring national food security and social stability. Therefore, grounded in the systematic perspective of sustainable development theory, this study carries significant strategic value. It aims to scientifically measure the agricultural sustainable development level (ASDL) in cities across Shandong Province and analyze its spatiotemporal evolution patterns and underlying driving factors. This research will facilitate the design of tailored local policies, advance high-quality agricultural transformation in Shandong, support the national agricultural green development strategy, and provide practical paradigms for selecting sustainability pathways in comparable regions.

While sustainable development has garnered significant attention, the existing literature shows considerable gaps in systematic and refined studies of the agricultural sector, hindering a deeper understanding of its sustainability mechanisms. First, a pronounced "sectoral focus ambiguity" exists in the research scope. A substantial body of research concentrates on sustainability assessments of urban economic systems or broad ecological systems. For instance, Yan et al. (2025) analyzed the economy-society-environment nexus, and Xu et al. (2021) examined the coordination between ecological quality and urbanization. Although systematic, these macro-level frameworks often overlook the unique constraints and specific development logic of agriculture. For example, Tomal (2020) study of rural-urban areas in Poland, while addressing rural development, integrates socioeconomic and infrastructural

dimensions without a strict focus on agricultural sector sustainability itself. This research bias results in insufficient depth in analyzing agriculture-specific resource and environmental pressures, technological path dependence, and policy needs. Second, a prevalent "macro-level preference" in the analytical scale neglects heterogeneity within provincial-level units. Most studies are conducted at macro levels, such as national (Yang et al., 2023), regional (Pang et al., 2024), or provincial scales (Liu et al., 2023). For instance, Huang et al. (2021) assessed ecological-economic coordination in Belt and Road countries, and Yin et al. (2023) analyzed the coordination among tourism, social welfare, and the ecological environment in Western China. While valuable for identifying broad trends, such research often relies on an implicit assumption of "regional homogeneity." This masks heterogeneity stemming from differences in resource endowments, economic structures, and development stages (Zhang et al., 2023), thereby hindering the formulation of targeted policies. Finally, a methodological imbalance exists, characterized by an "emphasis on assessment over explanation". Many studies focus on applying methods like the entropy weight method and coupling coordination degree models to depict spatiotemporal differentiation patterns (Qin et al., 2023). However, the exploration of underlying driving mechanisms often remains at the level of qualitative discussion or simple linear correlation (Yan et al., 2025). These studies often fail to uncover the complex, nonlinear effect pathways of various factors, as suggested by theories like the EKC, consequently limiting the precision of their policy implications.

To accurately characterize the spatiotemporal patterns of ASDL and explore its underlying drivers, this study employs an integrated and innovative methodological approach. The combined use of the entropy weight method and the coupling coordination degree (CCD) model establishes a reliable analytical foundation for measuring ASDL (He et al., 2024; Ge et al., 2023). Building on this foundation, the study further incorporates machine learning and interpretability techniques to enhance the analysis of driving factors. Specifically, we develop a "spatiotemporal assessment -mechanism interpretation" research framework, detailed in Fig. 1. First, the entropy weight-CCD model is used to measure the ASDL of individual cities in Shandong Province from 2013 to 2022. Second, kernel density estimation and GIS spatial analysis are applied to systematically examine the spatiotemporal evolution characteristics of ASDL. Finally, the study innovatively employs the integrated XGBoost-SHAP approach to overcome limitations of traditional econometric methods (Lundberg & Lee, 2017), facilitating a deeper analysis of the factors driving ASDL. This methodological combination allows the research to transcend simple correlation analysis and precisely identify the key drivers of ASDL and their operative mechanisms.

Building on the preceding analysis, this study aims to achieve the following two core research objectives: (1) To reveal the spatiotemporal evolution trend of ASDL in Shandong Province from 2013 to 2022; (2) To quantitatively analyze the key driving factors influencing ASDL and their nonlinear mechanisms. This study provides evidence-based support for agricultural policy in Shandong Province, while its methodological framework offers a valuable reference for sustainability research in comparable regions.

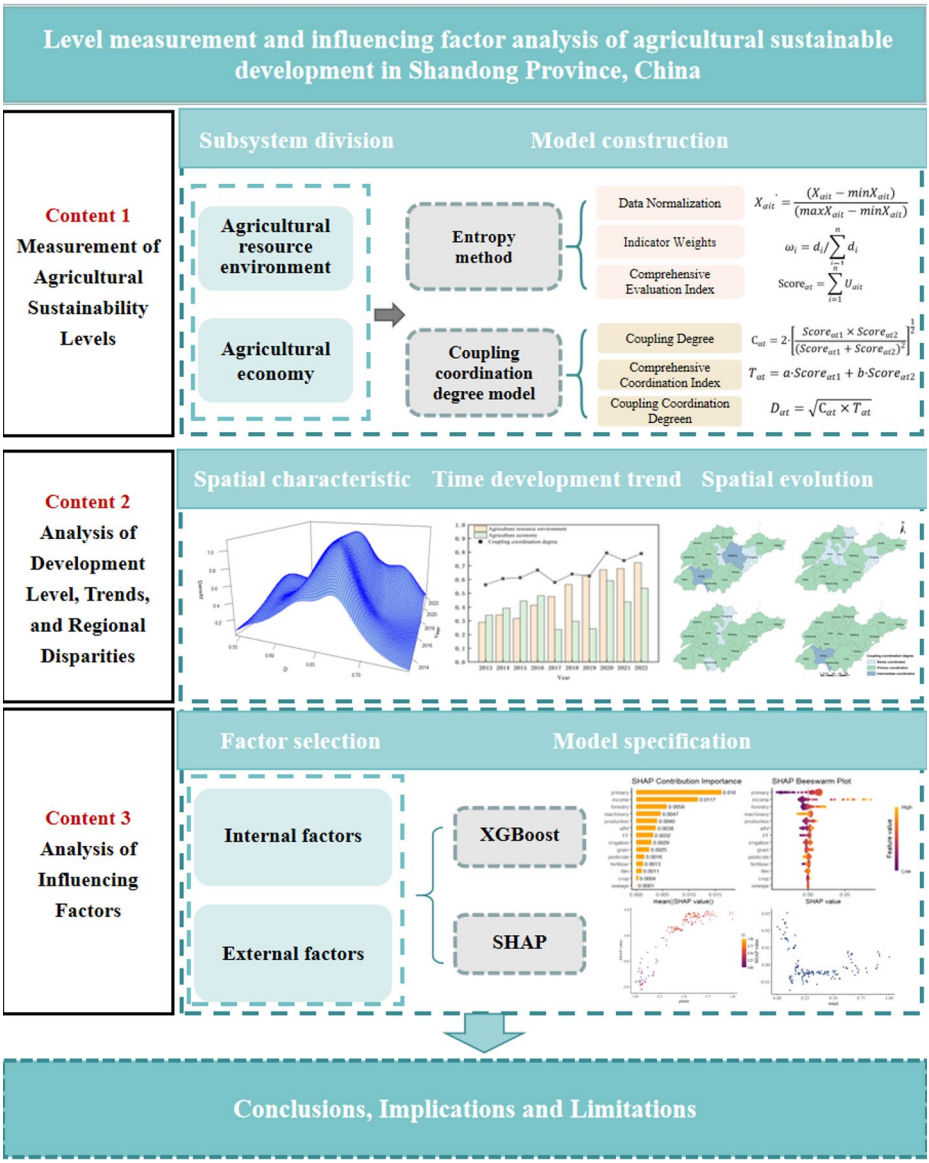

Fig. 1 Research framework of ASDL in Shandong Province

2 Materials and methods

2.1 Study region

Shandong Province is located on the eastern coast of China, downstream of the Yellow River (34° 22.9′–38° 24.01′ N and longitude 114° 47.5′–122° 42.3′ E). It spans 721.03 km latitudinally and 437.28 km longitudinally, with a land area of 155,800 km^2. In terms of eco-

nomic structure, agriculture dominates the province's economic structure: in 2022, Shandong's primary industry gross domestic product was 629.86 billion yuan, or about 7.2% of the national total. This large contribution to the national agricultural economy makes Shandong's agricultural sector an economic and social stabilizer.

From an ecological perspective, Shandong has undergone great improvements in ecological and environmental quality metrics related to air and river water quality. According to the 2022 Bulletin of Ecological and Environmental Conditions in Shandong, the average concentration of particulate matter smaller than 2.5 μm decreased by 63.3% in 2022 compared to 2013; during the same period, the proportion of days with good air quality increased by 33.2%, the number of heavily polluted days decreased from 60.8 to 3.5 (a decrease of up to 94.24%), and the chemical oxygen demand and ammonia-nitrogen levels in key rivers decreased by 27.7% and 75.8%, respectively.

2.2 Variables and data sources

2.2.1 Data sources and preprocessing

Data were primarily collected from the Shandong Statistical Yearbooks, supplemented by the ERA-Interim meteorological database from the European Centre for Medium-Range Weather Forecasts and geospatial data from the National Geomatics Center of China. We manually compiled these sources into a comprehensive panel dataset.

Laiwu City was administratively merged into Jinan City in 2018. To maintain consistent geographical boundaries throughout the study period, we excluded Laiwu City from our sample. We conducted a sensitivity analysis to assess this exclusion's potential impact by including Laiwu's data and comparing the results with our main specification. Results demonstrate that including or excluding Laiwu City did not significantly affect either the temporal trend of provincial ASDL or its spatial distribution patterns. This confirms the robustness of our findings.

2.2.2 Index system construction

To explore the dynamic evolution of ASDL in Shandong Province under the rural revitalization strategy, we selected 16 prefecture-level cities within the province as research subjects. Drawing on the work of previous scholars (Pang et al., 2024; Yin et al., 2023; Zhang et al., 2024), we established an indicator system with two secondary indicators: agricultural resource environment (ARE) and agricultural economic development (AED). The ARE system was assessed using 8 tertiary indicators, including resource endowment, resource utilization, environmental pollution, and governance. The AED system was examined through 6 tertiary indicators, focusing on rural residents' economic capacity and agricultural development level. Table 1 provides details.

2.2.3 Influencing factors

This study examines the driving mechanisms of ASDL in Shandong Province through factors selected from two dimensions: internal system characteristics and the external driving environment. Internal factors correspond directly to the tertiary indicators of the established

Table 1 Indicator system and weights for the ASDL

Secondary indicators	Third-level indicators	Variable abbreviation	Attribute	Indicator meaning	Weight
The ARE system	Multiple Cropping Index of Cultivated Land	*crop*	-	Crop Sowing Area/ Cultivated Land Area	0.1467
	Percentage of Effectively Irrigated Area to Cultivated Land Area (%)	*irrigation*	+	Effectively Irrigated Area/Cultivated Land Area	0.2094
	Total Mechanical Power per Unit Sowing Area (kw/hm²)	*machinery*	-		0.0864
	Total Afforestation Area	*forestry*	+		0.1028
	Fertilizer Application Intensity (t/hm²)	*fertilizer*	-	Pure Chemical Fertilizer Application Amount/Cultivated Land Area	0.1482
	Pesticide Application Intensity (t/hm²)	*pesticide*	-	Pesticide Application Amount/Cultivated Land Area	0.1097
	Agricultural Film Application Intensity (t/hm²)	*film*	-	Agricultural Film Application Amount/ Cultivated Land Area	0.0676
	Total Wastewater Discharge (10,000 tons)	*sewage*	-		0.1291
The AED system	Per Capita Disposable Income of Rural Residents (yuan)	*income*	+		0.3275
	Grain Yield per Unit Area (kg/hm²)	*grain*	+	Grain Output/Grain Sowing Area	0.1030
	Total Agricultural Output Value per Unit Sowing Area (100 million yuan/hm²)	*production*	+	Total Agricultural Output Value/Grain Sowing Area	0.1734
	Total Output Value of Agriculture, Forestry, Animal Husbandry, and Fisheries	*afhf*	+		0.1560
	Percentage of Primary Industry in GDP (%)	*primary*	+	Value Added of Primary Industry/Regional GDP	0.0951
	Fiscal Support for Agriculture	*FF*	+	Expenditure on Agriculture, Forestry, and Water Affairs/ General Public Budget Expenditure	0.1449

ASDL indicator system. These indicators represent the most direct manifestations of system sustainability.

External factors incorporate both human intervention intensity and geographical environmental endowment. For human intervention, selected factors are population density, fiscal revenue-to-GDP ratio, GDP per capita, and R&D intensity. These variables capture socioeconomic impacts on the agricultural system, reflecting resource and environmental pressure, macro-regulatory capacity, economic development stage, and technological innovation. For the geographical environment, factors include the ventilation coefficient and river density. The ventilation coefficient reflects the potential for atmospheric pollutant dispersion and removal, directly affecting gaseous pollution fluxes like nitrogen deposition.

River density indicates hydrological transport characteristics and the efficiency of non-point source pollutant aggregation and migration. Together, they characterize the region's inherent environmental capacity to assimilate agricultural pollution loads, providing a crucial supplement for understanding spatial heterogeneity in sustainable development levels under geographical constraints. Ventilation coefficient data were obtained from the ERA-Interim database of the European Centre for Medium-Range Weather Forecasts. River density data came from the National Geomatics Center of China. All other data were collected from city statistical yearbooks.

2.3 Method introduction

2.3.1 Entropy method

The entropy method serves as an objective weighting approach that quantifies the information value within datasets. Higher information content corresponds to reduced uncertainty and lower entropy values. Data processing was conducted using Stata 18.0, following established computational procedures.

Step 1, Data normalization.

Given the heterogeneity in indicator construction processes and the variations in dimensional units across indicators, it is necessary to first apply data standardization to eliminate these influences (Xu et al., 2023). The standardization is performed using the following formula:

① For positive indicators:

$$X'_{\alpha it} = \frac{(X_{\alpha it} - minX_{\alpha it})}{(maxX_{\alpha it} - minX_{\alpha it})} \tag{1}$$

② For negative indicators:

$$X'_{\alpha it} = \frac{(maxX_{\alpha it} - X_{\alpha it})}{(maxX_{\alpha it} - minX_{\alpha it})} \tag{2}$$

Where, $X_{\alpha\, it}$ and $X_{\alpha\, it}{}'$ represent the original data and normalized data for the indicator i of city α in the year t, respectively. The value range of the data after standardization is: $X_{it}{}' \in [0,1]$, $minX_{\alpha\, it}$ and $maxX_{\alpha\, it}$ represent the minimum and maximum values of the indicator i in the year t for city α, respectively.

Step 2, Weighting using the entropy method.

①Calculate the weight of the indicator i in the year t for city α:

$$\mathrm{Y}_{\alpha\mathrm{it}} = \mathrm{X}'_{\alpha\mathrm{it}} / \sum_{\alpha=1}^{\mathrm{m}} \sum_{\mathrm{t}=2013}^{2022} \mathrm{X}'_{\mathrm{itk}} \tag{3}$$

②Calculate the information entropy:

$$e_i = -\mathrm{K} \times \sum_{\alpha=1}^{m} \sum_{t=2013}^{2022} (Y_{\alpha it} \times lnY_{\alpha it}) \tag{4}$$

Where, $\mathrm{k} = \frac{1}{\ln(\mathrm{m} \times \mathrm{n})}$, m represents the number of cities, n represents the number of years, and $0 \leq e_i \leq 1$, When $\mathrm{Y}_{\alpha it=0,\ \mathrm{let}} \mathrm{Y}_{\alpha it} \times \ln \mathrm{Y}_{\alpha it} = 0$, let

③ Calculate the redundancy of information entropy:

$$d_i = 1 - e_i \tag{5}$$

④ Determination of indicator weights:

$$\omega_i = d_i / \sum_{i=1}^{n} d_i \tag{6}$$

⑤ Calculate the scores for each indicator and the comprehensive evaluation index for both systems:

$$\mathrm{U}_{\alpha it} = X'_{\alpha it} \times \omega_i \tag{7}$$

$$\mathrm{Score}_{\alpha t} = \sum_{i=1}^{n} U_{\alpha it} \tag{8}$$

Where, $U_{\alpha it}$ represents the entropy weight scores of the indicator i in the year t for city α, and $\mathrm{Score}_{\alpha t}$ represents the comprehensive evaluation index of the city α in the year t.

2.3.2 Coupling coordination degree model

Based on existing practices (Wang et al., 2023; Xiaohua et al., 2018), the coupling coordination degree is calculated after obtaining the scores for System 1 and System 2. The formulas are as follows:

Step 1, Calculation of coupling degree:

$$C_{\alpha t} = 2 \times \left[\frac{Score_{\alpha t1} \times Score_{\alpha t2}}{(Score_{\alpha t1} + Score_{\alpha t2})^2} \right]^{\frac{1}{2}} \tag{9}$$

Where, $Score_{\alpha t1}$ and $Score_{\alpha t2}$ represent the score of the city α in year t.

Step 2, Calculation of the comprehensive coordination index:

$$T_{\alpha t} = a \times Score_{\alpha t1} + b \times Score_{\alpha t2} \tag{10}$$

Where, T_{tk} represents the comprehensive coordination index of the city α in year t. Following existing practices, this paper sets $a = b = 0.5$.

Step 3, Calculation of coupling coordination degree:

$$D_{\alpha\,t} = \sqrt{C_{\alpha\,t} \times T_{\alpha\,t}} \tag{11}$$

Where, $D_{\alpha\,t}$ represents the coupling coordination degree between System 1 and System 2 of the city α in year t, and we use this index to characterize the level of sustainable agricultural development in Shandong Province. To more intuitively reflect the level of sustainable agricultural development, we adopt the coupling coordination classification criteria from relevant studies (Mondal et al., 2023; Tang et al., 2022) to categorize the agricultural sustainable development level, as shown in Table 2.

2.3.3 XGBoost model

To examine the complex mechanisms influencing ASDL, this study employs the XGBoost algorithm. Compared to traditional econometric approaches like Tobit or panel regression, XGBoost effectively captures complex nonlinear relationships and interaction effects among variables. This capability avoids limitations inherent in pre-specified functional forms (Shwartz-Ziv & Armon, 2022). Furthermore, compared to conventional machine learning methods, XGBoost incorporates regularization terms and second-order derivative information within its gradient boosting framework. These optimizations enhance prediction accuracy while controlling overfitting risk (Chen et al., 2016). The algorithm demonstrates robustness to missing values and automatically addresses multicollinearity. Therefore, XGBoost enables more flexible and robust identification of key driving factors and their mechanisms affecting ASDL in Shandong Province. The analysis was implemented in the R software, with relevant calculation formulas referenced from (Chen et al., 2016).

2.3.4 SHAP model

This study further applies the SHAP method to conduct an in-depth analysis of the XGBoost estimation results. SHAP interpretation resolves the opacity of machine learning models by quantifying both the magnitude and direction of each factor's contribution to agricultural sustainable development. This enables deeper investigation of driving factors. Furthermore, SHAP provides sample-level decomposition, revealing feature contributions and interaction

Table 2 The standard for dividing the degree of coupling coordination

Grade	Coordination degree interval	Coupling coordination degree	Grade	Coordination degree interval	Coupling coordination degree
1	0< D≤0.1	Extremely disordered	6	0.5< D≤0.6	Barely coordinated
2	0.1< D≤0.2	Severely disordered	7	0.6< D≤0.7	Primary coordination
3	0.2< D≤0.3	Moderately disordered	8	0.7< D≤0.8	Intermediate coordination
4	0.3< D≤0.4	Mildly disordered	9	0.8< D≤0.9	Good coordination
5	0.4< D≤0.5	On the verge of being disordered	10	0.9< D≤1	Quality coordination

effects that clarify each feature's role in ASDL variation. The implementation was implemented in R software, with relevant calculation formulas referenced from Lundberg and Lee (2017).

3 Empirical results and discussion

3.1 Analysis of agricultural sustainable development

3.1.1 Time development trend

First, this study employs the R Studio to generate a three-dimensional kernel density plot, observing the evolution characteristics of the overall ASDL in Shandong Province over time (as shown in Fig. 2). The three-dimensional kernel density estimation integrates spatial visualization with nonparametric density approximation, quantitatively revealing distribution patterns, polarization dynamics, and regional disparities in the ASDL across Shandong Province. Analysis reveals that the ASDL predominantly clusters within the 0.6–0.65 range. The distribution centroid demonstrates rightward migration accompanied by vertical elevation, suggesting progressive enhancement in the ASDL. In terms of distribution shape, there is a clear trend of shifting from left-skewed to right-skewed, showing that over time, more cities are emerging with higher ASDL.

Moreover, during 2013–2022, most years show a bimodal distribution, and the span between the two peaks gradually widens, indicating that there is a multi-polar divergence in the ASDL. This phenomenon stems from the combined effect of policy pilot mechanisms and initial regional endowments. Shandong Province implemented policies like *Agricultural Modernization* and the *National Agricultural Green Development Pilot Zone* through a "pilot-first" strategy, leveraging demonstration areas to drive regional progress.

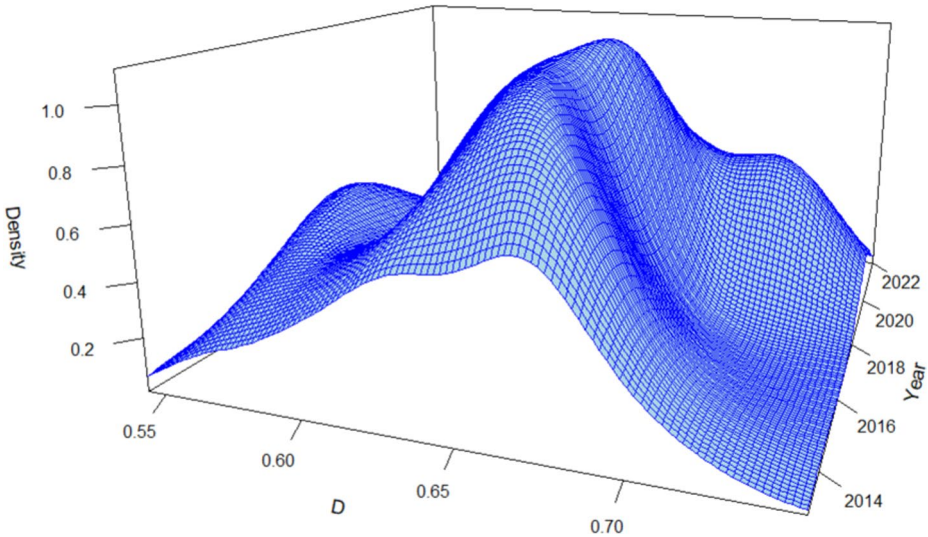

Fig. 2 Three-dimensional kernel density map of the ASDL

This approach allowed well-endowed cities with stronger foundations to secure preferential policy support and resources more readily. This created a "Matthew effect," accelerating their transformation. In contrast, cities with weaker foundations struggled to secure comparable support, facing greater transitional challenges. Consequently, the development gap between these two groups widened, which is directly manifested in the expanding peak span of the kernel density curve. Without targeted policy interventions, this disparity risks becoming entrenched or even widening, thereby hindering coordinated green development across the province.

Second, by integrating the data of various indicators, we obtained the time-series characteristics of the development of the ARE and AED systems, as well as the ASDL in Shandong Province, as shown in Fig. 3. Except for a slight decline in a few years, the ASDL in Shandong Province shows an overall upward trend. According to the development level and type, the agricultural sustainable development process in Shandong Province can be divided into three stages.

The first stage occurred from 2013 to 2016. During this period, the ASDL showed a significant upward trend, rising from 0.561 in 2013 to 0.668 in 2016, indicating a continuous increase. Additionally, the development level of the AED system was consistently higher than that of the ARE system, indicating a lag in the environmental system. The ASDL rose

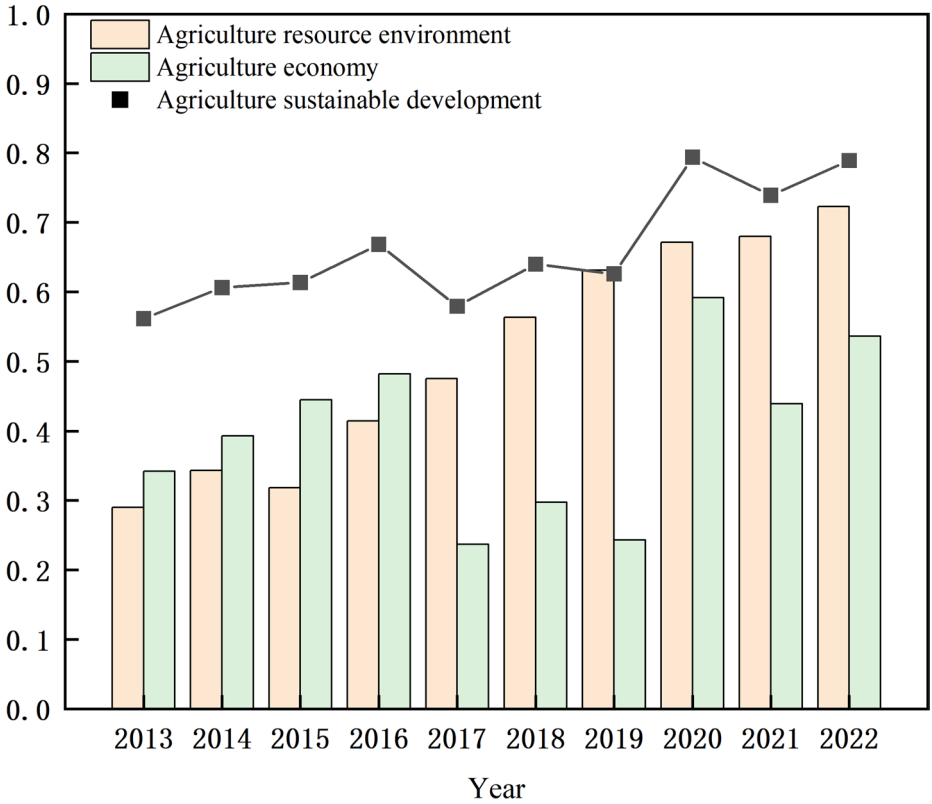

Fig. 3 Time trend diagram of two systems and the ASDL in Shandong Province

steadily from 0.342 in 2013 to 0.482 in 2016. In contrast, the ARE level declined slightly in 2014 but otherwise trended upward evidently.

The second stage occurred from 2017 to 2019. During this period, a significant decline occurred in the ASDL of Shandong Province, which remained at a low level of around 0.62 in the following two years. It can also be observed that the ARE development level, which was below that of AED in the previous stage, has now surpassed it, indicating a shift from an ecological lag type to an economic lag type. In this stage, the ARE development showed a sustained upward trend with a higher growth efficiency than in the first stage. However, the AED, affected by previous cumulative factors, was not optimistic and experienced notable fluctuations. In the later period, through appropriate policy adjustments and changes in the development model, it is expected that the coordinated development of resources, environment, and economy can be achieved.

The third stage occurred from 2020 to 2022. The ASDL in the province has increased significantly and stabilized between 0.7 and 0.8. In 2020, it reached 0.794, the highest level in history. In 2021, the coupling relationship between the two systems declined due to a slowdown in the ARE and a notable decline in the AED; overall, economic development lagged behind. This fluctuation is largely attributable to the impact of the COVID-19 pandemic. As the pandemic spread nationwide, strict containment measures substantially disrupted production and trade activities. *The Progress towards the Sustainable Development Goals* (2022) report notes that the pandemic caused "delays in production inputs," severely disrupting supply chains and global trade while triggering a surge in food prices. Within the agricultural sector, material and labor shortages, financial pressures, blocked sales channels, and price volatility significantly heightened risks for green production. These challenges directly adversely affected the production and marketing of agricultural products and by-products (Mao et al., 2023).

3.1.2 Spatial characteristic manifestations

Analysis of city-level ASDL changes from 2013 to 2022 in Table 3 shows that most of the 16 cities in Shandong Province experienced improvement during the study period. Meanwhile, substantial shifts occurred in city rankings based on ASDL. This paper examines the ranking patterns in 2013 and 2022 for a detailed comparison. In 2013, the top three cities in terms of the ASDL were Weifang, Jining, and Binzhou. In 2022, the top three were Jining, Tai'an, and Weihai. Seven cities, including Binzhou, Dezhou, Dongying, Jinan, Linyi, Weifang, and Zaozhuang, experienced a decline in their ranking order. The ranking order of the remaining cities remained unchanged or improved.

The observed dynamics in city rankings likely stem from variations in resource endowments and development pathways across the province. Jining, a major grain-producing area on the Luxi Plain, likely achieved steady improvement through its intensive and standardized agricultural production. In contrast, Tai'an and Weihai capitalized on distinct comparative advantages. Tai'an benefits from a green agriculture orientation leveraging Mount Tai's ecological barrier, while Weihai developed a refined and efficient model suited to its coastal context. Conversely, Weifang's highly industrialized agricultural model led to mounting resource and environmental pressures, particularly from water constraints. Cities including Binzhou, Dongying, and Jinan appear affected by crowding-out effects from rapid industrialization and urbanization. Overall, ASDL in Shandong Province showed consistent

Table 3 Changes in the score, ranking, and rank of the ASDL in 2013 and 2022

Municipal	2013		2022		Position order change
	Agricultural sustainable development level	Ranking	Agricultural sustainable development level	Ranking	
Binzhou	0.6836	3	0.6504	8	−5
Dezhou	0.6747	5	0.6386	10	−5
Dongying	0.6204	13	0.6079	14	−1
Heze	0.6291	11	0.6369	11	0
Jinan	0.6019	14	0.6033	16	−2
Jining	0.7058	2	0.7065	1	1
Liaocheng	0.6638	8	0.6712	5	3
Linyi	0.6789	4	0.6534	6	−2
Qingdao	0.5905	15	0.6228	12	3
Rizhao	0.6403	9	0.6404	9	0
Tai'an	0.6728	6	0.6955	2	4
Weihai	0.6355	10	0.6869	3	7
Weifang	0.7062	1	0.6738	4	−3
Yantai	0.6700	7	0.6526	7	0
Zaozhuang	0.6272	12	0.6061	15	−3
Zibo	0.5612	16	0.6108	13	3

improvement. The number of cities attaining at least primary coordination increased from 14 in 2013 to all 16 cities by 2022.

Based on the GIS analysis software, spatial evolution maps of the ASDL were constructed using data from four cross-sectional points (2013, 2016, 2019, and 2022) across prefecture-level cities, as shown in Fig. 4.

In 2013, most cities in Shandong Province were at the primary coordination stage. Only Zibo and Qingdao remained at the barely coordinated stage, whereas Weifang and Jining had achieved intermediate coordination. As core areas of the "Blue-Yellow" Strategy, Zibo and Qingdao showed strong industrial orientation toward the secondary sector, which accounted for 56.8% and 45.5% of their respective economies in 2013. The concentration of resources in industry and services, alongside environmental pressures from industrialization, constrained their ASDL. In contrast, Weifang and Jining leveraged superior agricultural resource endowments. Weifang contained approximately 10% of the province's cultivated land, while Jining possessed a favorable eco-agricultural foundation around Nansi Lake. Supported by early agricultural industrialization policies, both cities achieved remarkable results in product processing and eco-agriculture, thereby enhancing their ASDL.

By 2016, the provincial ASDL had deteriorated. Jinan and Dongying declined from primary to barely coordinated, while Weifang and Jining dropped from intermediate to primary coordination. This decline coincided with a period of rapid industrialization and urbanization. In Dongying, a petroleum resource-based city, oil extraction and processing constituted over 60% of large-scale industrial value-added in 2016. This high-intensity energy development squeezed agricultural space and damaged regional water and soil resources due to its high energy consumption and emissions. Meanwhile, Jinan, as the provincial capital, experienced strengthened agglomeration effects, leading to further concentration of fiscal and ecological resources in the central urban area. This concentration relatively

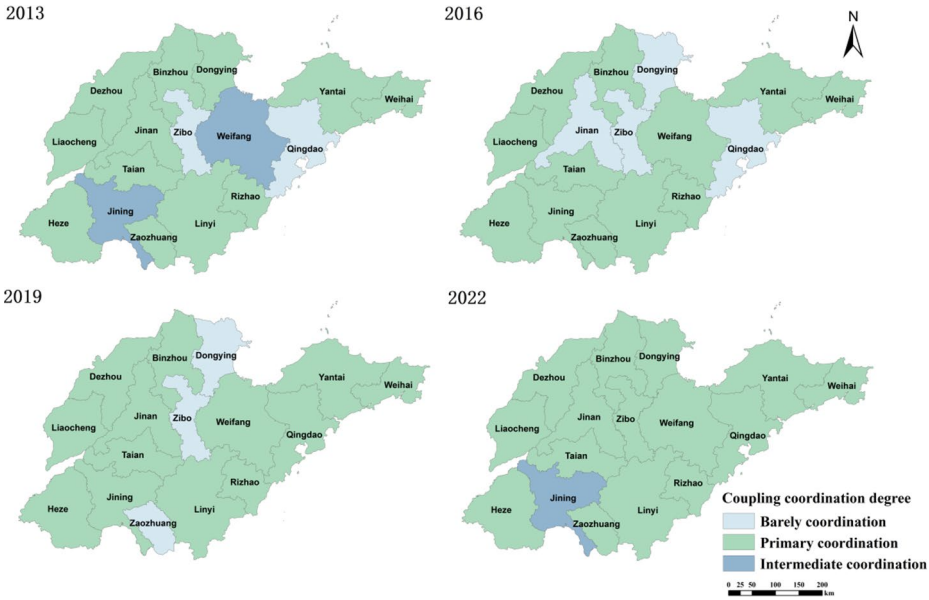

Fig. 4 Spatial evolution of the ASDL

reduced investment in agriculture and ecological protection, resulting in declining agricultural sustainability.

By 2019, Jinan and Qingdao had improved from barely coordinated to primary coordination. This progress stemmed from substantive efforts in industrial restructuring, innovation-driven development, and ecological governance. From 2018, Jinan utilized the Old and New Growth Driver Conversion Pilot Zone to phase out outdated capacity and develop smart agriculture and ecological recycling technologies, significantly reducing energy consumption per unit of agricultural value-added. Qingdao, leveraging its Marine Economy Development Demonstration Zone, focused on developing green sectors like modern fisheries and agricultural technology services, thereby upgrading traditional agricultural models.

By 2022, all cities had reached at least the primary coordination stage, with Jining returning to the intermediate coordination stage. This improvement demonstrates the effectiveness of Shandong's regional coordinated development strategy and rural revitalization practices. Through the "Jinan Metropolitan Circle" development orientation and initiatives like Weifang's "Qilu Model of Rural Revitalization," the region enhanced infrastructure connectivity, joint ecological construction, environmental governance, and agricultural modernization. These measures collectively drove a province-wide leap in ASDL coordination.

3.2 Influencing factor analysis based on the XGBoost + SHAP model

This study used 10-fold cross-validation with early stopping to optimize XGBoost hyperparameters. This approach assessed model generalization performance and identified the optimal parameter set. Hyperparameter tuning results indicate that the model achieved optimal performance with a learning rate of 0.08 and 59 training rounds. With this configuration, 10-fold cross-validation produced a root mean square error of 0.0323 ± 0.0096, while the

R^2 value for each fold exceeded 0.85. These metrics demonstrate strong predictive accuracy and model stability, providing a reliable basis for subsequent driver analysis. Based on these results, the study systematically analyzes SHAP results for both internal and external factors.

3.2.1 Internal factors

Using the XGBoost-SHAP model, this study first identifies the core internal factors influencing ASDL and their operational mechanisms (Fig. 5). The SHAP summary plot identifies four predominant factors: primary industry share (*primary*), rural residents' disposable income (*income*), afforested area (*forestry*), and total mechanical power per unit sown area (*machinery*). This ranking reflects a core rationale where regional industrial structure, rural household well-being, ecological capital stock, and technological application patterns constitute the four fundamental dimensions driving ASDL in Shandong Province.

These key factors influence agricultural sustainability through distinct pathways. A higher primary industry share indicates positive effects from agricultural specialization and resource concentration. In this major agricultural province, such a share signifies the central economic role of agriculture, which channels policy resources, fiscal investment, and technical services into the sector. This fosters intensive land management and industrial chain integration, ultimately enhancing system-level coordination. Rising disposable income for rural residents establishes a microeconomic foundation for sustainable development. It directly improves farmer welfare and enables crucial capital accumulation for agriculture's green transformation. Higher-income households are more willing and able to invest in cost-effective, long-term conservation measures like water-saving irrigation, organic fertilizers, and green pest control. This creates a virtuous cycle where income growth stimulates green investment, leading to environmental improvement and enhanced ASDL (Liu et al., 2020). Afforestation expansion directly enhances agricultural ecosystem services. As critical ecological assets, forests provide natural "ecological insurance" to farmland by conserving water, retaining soil, regulating climate, and protecting biodiversity. Furthermore, market mechanisms like forestry carbon sinks and ecological compensation translate these ecological benefits into economic value (Wang et al., 2018), providing new impetus for ASDL. The

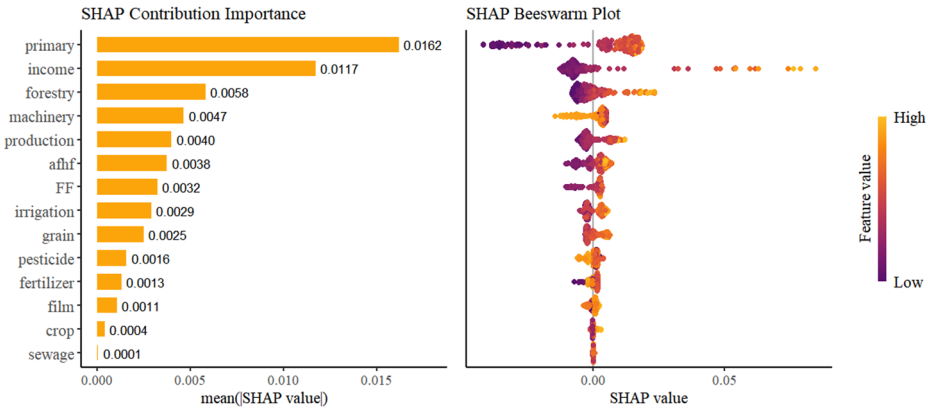

Fig. 5 The importance ranking and impact of internal factors. Note: The abbreviations of the variables in the figure can be found in Table 1

negative impact of mechanical power per unit area highlights environmental costs linked to prevailing technology use. The predominant mechanization model in Shandong often relies on traditional, high-consumption, high-horsepower machinery. This approach can cause soil compaction, degrade soil structure, and generate significant diesel consumption and emissions. This finding cautions that merely increasing the mechanization rate does not ensure sustainable development. Its positive potential requires transitioning towards precision, intelligent, and low-carbon machinery, integrated with green agronomic practices like conservation tillage.

Factors including agricultural plastic film (*film*) use, multiple cropping index (*crop*), and wastewater discharge (*sewage*) exhibited relatively low SHAP values. This does not imply they are harmless. A more plausible explanation is that regulatory controls in Shandong had mitigated their most severe impacts during the study period. Consequently, their inter-city variation was insufficient to establish them as dominant explanatory variables.

This study further examines interaction effects using SHAP, acknowledging that influences on ASDL are not simply additive and may involve synergies. The results indicate synergistic effects among certain driving factors, with the interaction between the share of the primary industry and the intensity of fiscal support (*FF*) for agriculture being the most significant. As shown in Fig. 6, ASDL increases substantially when both primary industry share and agricultural fiscal support are high. This suggests that agricultural green transformation depends not only on industrial structure but also on complementary institutional support. In agriculture-dominated regions, the sector's prominence offers clear policy targets and efficient resource allocation channels for public finance. Under these conditions, fiscal funds systematically overcome resource and environmental constraints through infrastructure modernization, green technology incentives, and ecological compensation. This effectively transforms industrial structure advantages into sustainable development momentum, enhancing both resource utilization efficiency and environmental sustainability of the agricultural system.

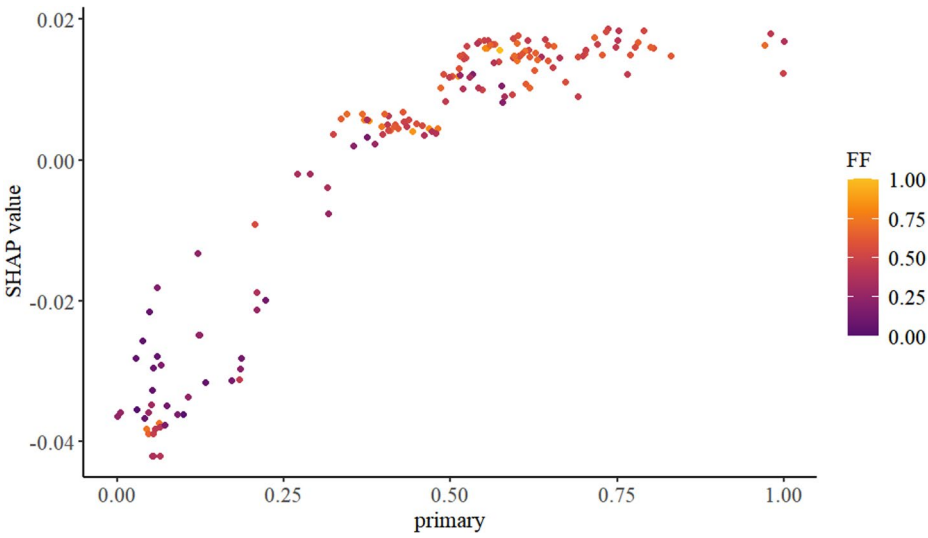

Fig. 6 The interaction between the share of the primary industry and the intensity of fiscal support

3.2.2 External factors

Analysis of external factors identifies four core drivers influencing agricultural sustainable development: population density (*popul*), GDP per capita (*pergdp*), fiscal revenue scale (*finance*), and ventilation coefficient (*wind*) (Fig. 7). These factors collectively influence system sustainability through both synergistic and trade-off relationships.

GDP per capita shows a significant negative relationship with ASDL, consistent with findings by Pang et al.(2024). This indicates that economic growth in Shandong Province remains partially dependent on resource-intensive development patterns. Rapid industrialization and urbanization intensify agricultural ecological pressures through resource competition and pollution spillovers, aligning with the ascending phase of the EKC theory. In contrast, fiscal revenue scale exhibits a positive effect, underscoring government fiscal capacity's role in reconciling economic growth with environmental protection. Greater fiscal revenue mitigates economic growth's negative externalities by funding agricultural infrastructure, ecological compensation, and modernization, creating a compensatory effect (Liu et al., 2020).

The SHAP value distributions for population density and ventilation coefficient show high dispersion, indicating complex nonlinear mechanisms in their effects on agricultural sustainability. This study employed SHAP dependence plots to further examine these two factors. Figure 8 reveals clear threshold effects in both population density and ventilation coefficient impacts. Population density follows an inverted U-shaped relationship. Below 760 persons/km^2, population agglomeration expands labor supply and market demand, fostering agricultural intensification and technological progress. However, exceeding this threshold triggers resource constraints and ecological pressures that progressively undermine agricultural system sustainability. This pattern aligns with the development stage theory. Conversely, the ventilation coefficient shows a U-shaped pattern. Below the threshold of 1387.9, increased ventilation enhances the transport of atmospheric pollutants from upwind industrial and urban areas. This leads to soil acidification and crop damage through dry/wet deposition, impairing agricultural ecological health (Fowler et al., 2009). Above this threshold, strong diffusion capacity dominates environmental processes. This reduces

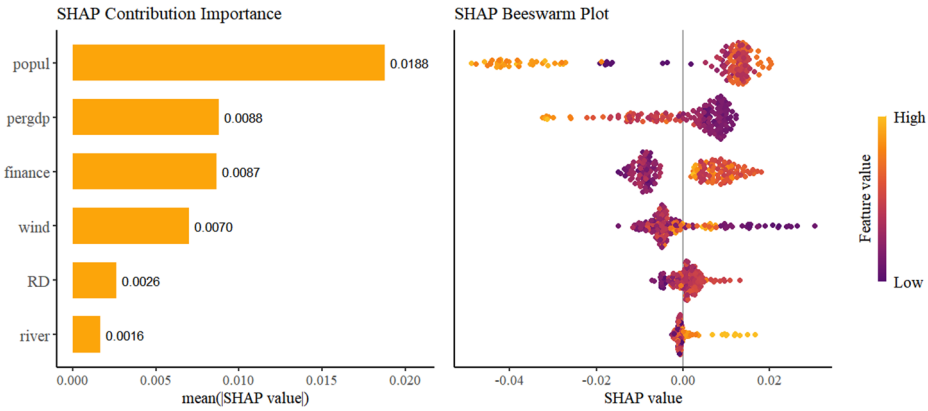

Fig. 7 The importance ranking and impact of external factors. Note: *popul* represents population density; *pergdp* represents per capita GDP; *finance* represents fiscal revenue; *wind* represents ventilation coefficient; *RD* represents R&D intensity; *river* represents river density

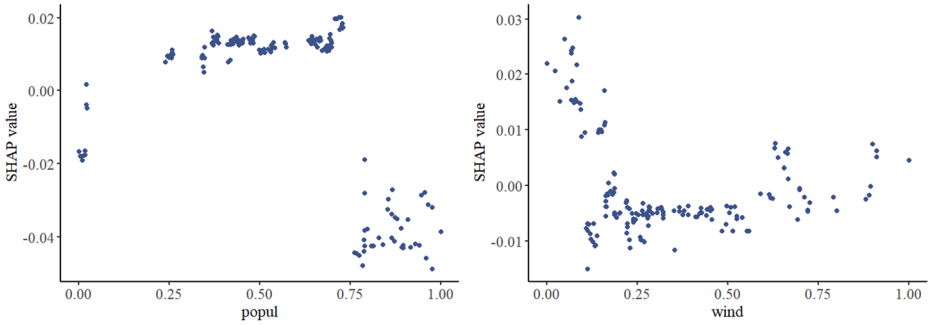

Fig. 8 The threshold effects of population density and ventilation coefficient. Note: The back-transformed thresholds from the model indicate a population density threshold of 760 persons/km^2 and a ventilation coefficient threshold of 1387.9

industrial pollutant deposition and dilutes agricultural emissions like ammonia, thereby alleviating environmental stress (Wang et al., 2018).

4 Conclusions and implications

4.1 Conclusions

This paper combines the CCD model with the XGBoost-SHAP framework to evaluate the change of agricultural sustainable development level (ASDL) in Shandong Province from 2013 to 2022, as well as the effects of internal and external factors. The following conclusions are drawn:

(1) In terms of temporal trends, the ASDL in Shandong Province increased from 0.561 to 0.789, indicating an overall upward trajectory. It progressed from barely coordinated to intermediately coordinated, albeit with fluctuations. The development type shifted from ecologically lagging to economically lagging around 2016.
(2) Regarding spatial evolution, the rankings of ASDL among the 16 prefecture-level cities shifted significantly. The top three cities shifted from Weifang, Jining, and Binzhou in 2013 to Jining, Tai'an, and Weihai in 2022. Meanwhile, inter-city disparities widened, even as most cities showed improvement in their agricultural sustainable development levels.
(3) Analysis based on the XGBoost-SHAP framework identifies key drivers of ASDL in Shandong Province. Among internal factors, the primary industry share, rural residents' disposable income, afforested area, and mechanical power per unit sown area are most influential. The primary industry share, income, and afforested area positively affect ASDL, whereas mechanical power has a negative effect. Fiscal support for agriculture also shows a synergistic effect with the primary industry share. For external factors, population density, GDP per capita, fiscal revenue scale, and ventilation coefficient are most significant. Among these, GDP per capita negatively affects ASDL, fiscal revenue scale has a positive effect, population density follows an inverted U-shaped pattern, and the ventilation coefficient shows a U-shaped relationship.

The findings of this study can provide empirical support for sustainable agricultural development in China. As a major agricultural country, China's progress in enhancing agricultural sustainability holds significant implications for achieving national food security and rural revitalization strategic objectives. Furthermore, the key factors identified in this research contribute empirical evidence from China to the global discourse on agricultural sustainability.

4.2 Policy implication

(1) Implement differentiated regulatory strategies to address entrenched disparities in ASDL through region-specific governance approaches. For ecologically lagging regions, establish special agricultural ecological compensation funds prioritizing saline-alkali land improvement in key areas, including Dongying, coastal Binzhou, southern Weifang, and Laizhou in Yantai. Simultaneously, enhance regulatory efforts to promote cleaner industrial transformation in Dezhou and Zibo. For economically lagging regions, policies should enhance value by leveraging agricultural industrialization advantages in areas like Weifang and Linyi. This includes establishing deep-processing parks and promoting integrated "leading enterprise + cooperative" models to improve processing conversion rates and product added value.

(2) Building on SHAP analysis of internal drivers, policies should strengthen targeted investments and coordination in crucial agricultural sectors. Specifically, support agricultural development through training for rural tertiary integration and foster new business models like e-commerce and leisure agriculture. These measures can increase agricultural value-added and diversify farmers' income sources. Promote green machinery replacement through special subsidies for electrifying high-energy-consumption equipment, mitigating soil compaction, and emissions. Furthermore, in agriculturally dominant regions, fiscal support should be strengthened and prioritized for green production technologies, ecological compensation, and infrastructure development, maximizing policy leverage through synergistic effects.

(3) Based on SHAP analysis of external factors, strictly enforce industrial and environmental negative lists by strengthening oversight to restrict high-pollution industries in agricultural zones. This will alleviate negative environmental externalities and resource competition from rapid industrialization. Concurrently, establish cross-regional agricultural ecological compensation funds to offset industrialization's spillover effects on agriculture. For cities approaching or exceeding 760 persons/km^2, strictly control development intensity and implement quality-oriented population and industrial policies to prevent resource overload. Where the standardized ventilation coefficient falls below 1387.9, poor atmospheric dispersion causes deposition of local and upwind industrial pollutants, intensifying environmental pressure. Prioritize these areas for coordinated air pollution control zones with stricter industrial emission standards, reducing cross-media pollutant risks to agricultural ecosystems.

4.3 Limitations and future research

Notwithstanding Shandong Province's leading role in China's agricultural development and its diverse topography—encompassing plains, hills, and coastal areas—along with varied agricultural systems, which make it a representative and insightful case for comparative analysis, significant disparities persist across different regions in terms of natural conditions, crop structures, and even cultural practices. Therefore, policy implications derived from this context may have limitations in broader application, and policy formulation must be tailored to local realities. Future research could extend to wider geographical areas, establishing inter-provincial comparisons. Furthermore, a top-down analytical framework based on sustainable development models, such as DPSIR or SLF, is well-suited for such studies, as it facilitates the examination of pressure-response mechanisms within systems through causal pathways.

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Data availability The data presented in this study are available on request from the authors, upon reasonable request.

Declarations

Informed consent statement Not applicable.

Institutional review board statement Not applicable.

Conflict of interest The authors declare no conflict of interest.

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